

Sky Shih-Kai Hong

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EDUCATION

- **National Yang Ming Chiao Tung University (NYCU)** Hsinchu, Taiwan
M.S. in Data Science and Engineering (IDS&E); Advisor: Prof. Grace Yu-Chun Yen *Jun 2025 – Expected 2027*
 - **Publication (R&R):** Second author on an ACM CHI 2026 full paper in revise-and-resubmit; contributed to research iteration including study design and system prototyping.
 - **Teaching / Mentoring (Capstone):** TA for *Multimedia and Human-Computer Interaction Capstone* (Prof. Yen, Prof. Chang); mentored 40+ students across 8 teams with ongoing technical/design reviews.
- **National Taiwan Normal University (NTNU)** Taipei, Taiwan
B.S. in Computer Science and Information Engineering (CSIE) *Sep 2021 – Jun 2025*
 - **Honors & academics:** Outstanding Student Award (AY112; 1/department); GPA 3.82; 159 credits; EMI-certified (E1, 18 EMI credits); Educational Program for Secondary School Teachers (IT).

EXPERIENCE

- **Institute of Information Science, Academia Sinica (Hinagiku Team)** Taipei, Taiwan
Research Grant Student, Project Manager & Researcher — Prof. Ling-Jyh Chen *Feb 2025 – Jun 2025*
 - **Leadership & execution:** Led 6+ developers and drove delivery via 60+ issues and 200+ PRs; owned roadmap, task breakdown, and issue board operations.
 - **Stakeholder-driven iteration:** Ran 10+ sessions with teachers/stakeholders; validated Think-Pair-Share (TPS) end-to-end and translated classroom constraints into shippable features.
 - **System & stack:** Real-time discussion support with voice transcription/analysis; SvelteKit, Firebase (Firestore/Auth) on GCP, GCS/Cloudflare R2, Genkit LLM (multi-provider). GitHub | Wiki
- **National Taiwan Normal University (NTNU)** Taipei, Taiwan
Teaching Assistant, Programming I & II (C) — Prof. Po-Wen Chi *Sep 2023 – Jun 2024*
 - **Teaching excellence at scale:** Supported 80–120 students in C programming (pointers, strings, structs, memory); earned 4.9/5 student rating (highest among 5 TAs) and received a university-wide Outstanding TA award; co-ran 11 optional TA Hours (230+ attendances).
 - **Automation & assessment:** Authored exam/assignment problems and built Bash/Makefile-based autograding pipelines to standardize builds, run test cases, and streamline grading at scale.

PROJECTS

- **UniCourse — Cross-University Course Search Database (Open Data):** Built a course search platform; implemented data collection/normalization and delivered a searchable UI. Presented at COSCUP 2023. (Talk)
- **LLM Preference Classification — Feature Engineering & Evaluation Pipeline:** Built an explainable feature dataset (57,477 A/B pairs, 28 features) and designed end-to-end experiments/metrics; feature-based model achieved 43.61% accuracy and 0.605 Macro-AUC, outperforming TF-IDF baselines.

ACHIEVEMENTS AND LEADERSHIP

- **Macronix Science Award (19th) — Excellence Award (Top 20/661):** Awarded for academic impact.
- **Founder & Lead, Google Developer Student Club (GDSC) NTNU:** Led a 24-member team; organized study groups, student-led courses, and industry exchange events with 1000+ total attendances in the first year.
- **Network Administration & System Administration (NTU CSIE) — Hands-on Infra Labs:** Placed 1st in the final practical exam among 21 teams; built VMs (virsh), configured firewall rules, DNS, NFS, and LDAP-based account management.
- **Legacy Server Modernization — Ubuntu/WordPress Maintenance:** Upgraded a 7+ year unmaintained Ubuntu + WordPress server; implemented backups, SSL issuance, and performance tuning; deployed/maintained sites with Cloudflare and Oracle Cloud.

SKILLS

- **Languages:** : Python, C/C++, TypeScript/JavaScript
- **Systems:** : Linux/Unix, Git, Bash, Makefile, GitHub Actions, REST APIs, basic networking
- **Cloud/Web:** : GCP, Oracle Cloud, Docker, Firebase (Firestore/Auth), SvelteKit, Cloudflare R2, Genkit